KAI JOSHUA CAPEWELL

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GitHub: github.com/KaiCapewell1

PROFESSIONAL PROFILE

Computer Science student specializing in gameplay engineering and real-time systems programming. Experienced in building modular C# game architectures, Finite State Machine AI, and asynchronous scene loading in Unity. Actively expanding technical capability in C++ and Unreal Engine. Seeking a 12-month software or gameplay engineering placement starting June 2027.

TECHNICAL SKILLS

- **Languages:** C#, C++ (learning), Python, PHP, JavaScript, SQL
- **Game Development:** Unity (Advanced), Unreal Engine (Intermediate), Object-Oriented Design, Finite State Machines, Vector Math
- **Tools & Databases:** Git, GitHub, MySQL, Oracle
- **Systems & Scripting:** Arch Linux (Bash scripting, CLI tooling), Windows, Networking Fundamentals

PROJECTS & EXPERIENCE

(All of my projects can be found on https://kicruft.online)

2D RPG Gameplay Systems | Unity, C#

- Engineered state-driven enemy AI (Patrol, Chase, Combat) using Finite State Machines and vector line-of-sight checks.
- Built responsive player movement, collision handling, and interactive gameplay mechanics in C#.
- Designed event-driven UI, dialogue progression systems, and modular state transitions.
- Implemented asynchronous scene loading using Unity SceneManagement for seamless area transitions.
- Maintained clean version control and codebase scalability using Git with structured feature-branching workflows.

Personal Portfolio Website | React, JavaScript

- Built and deployed a responsive React portfolio to showcase technical gameplay demos and code repositories.

- Developed modular, reusable UI components optimized for desktop and mobile layouts.

Linux Environment Setup | Personal Project

- Configured a custom Arch Linux development environment optimized for systems programming and workflow automation.
- Scripted system administration tasks and developer tooling using Bash and CLI commands.

EDUCATION

BSc (Hons) Computer Science

Aston University | *2025 – 2028 (Expected)*

Level 4 Computing and Digital HNC

Walsall College | *2024 – 2025*

Level 3 Computing and Digital Extended Diploma

Walsall College | *2022 – 2024*

- **Grade:** Merit, Merit, Merit (MMM)
- **Key Modules:** Software Design, Databases, Networking, Cyber Security, Systems Architecture